

Part 2: Frequency Method – Dynamic Determination of k

Mass m (kg)	Frequency f (Hz)	$\omega = 2\pi f$ (rad/s)	ω^2 (rad ² /s ²)	k = m ω^2 (N/m)
0.05	3.6154	22.71622816	516.0270218	25.80135109
0.1	2.5285	15.88703405	252.3978509	25.23978509
0.15	2.1538	13.53272451	183.1346328	27.47019492

Mean k (N/m)	26.1704437
Std dev k	1.160109485
Standard error	0.669789524
k $\pm$ SE (N/m)	26.17 $\pm$ 0.67

Comparison w Part 1 **Good agreement (<10% diff)**